

Relative Acidities of Chlorine-substituted Carboxylic Acids (Cambridge (CIE) A Level Chemistry): Revision Note

Relative Acidities of Chlorine-Substituted Carboxylic Acids

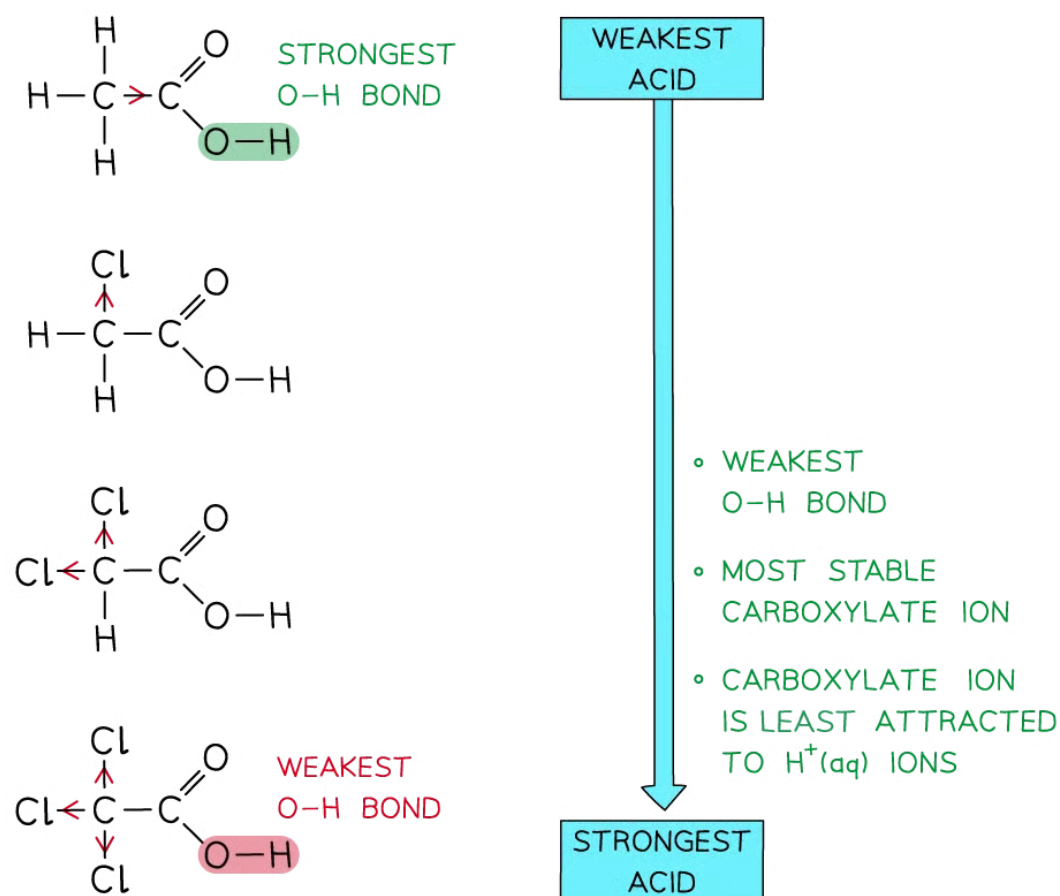
- **Electron-withdrawing** groups bonded to the carbon attached to the -COOH group make the carboxylic acids **stronger acids**
- This is because the O-H bond in the **undissociated acid molecule** is even further weakened as the electron-withdrawing group draws even more electron density away from this bond
- Furthermore, the electron-withdrawing groups extend the **delocalisation** of the negative charge on the -COO^- group of the carboxylate ion
- The -COO^- group is now even more stabilised and is less likely to bond with an H^+ ion
- **Chlorine-substituted carboxylic acids** are examples of carboxylic acids with electron-withdrawing groups

pK_a values of ethanoic acid and chlorine-substituted derivatives table

Acid	pK_a at 25 °C
Ethanoic acid, CH_3COOH	4.8
Chloroethanoic acid, CH_2ClCOOH	2.9
Dichloroethanoic acid, CHCl_2COOH	1.3
Trichloroethanoic acid, CCl_3COOH	0.6

- The pK_a values of ethanoic acid and **chloro-substituted derivatives** show that the **more** electron-withdrawing groups there are on the carbon attached to the -COOH group, the **stronger** the acid

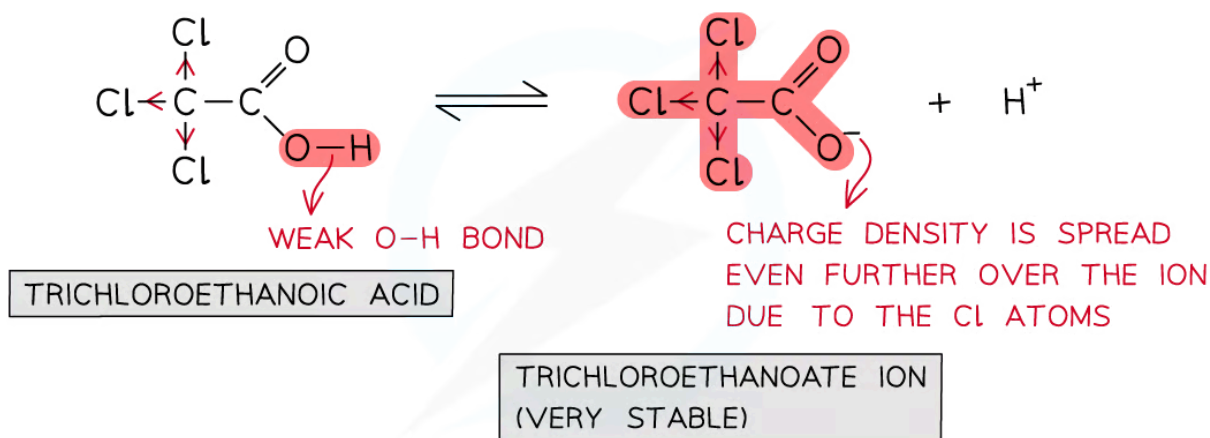
Comparing the relative acidities of chlorine substituted derivatives of ethanoic acid



The more chlorine atoms there are in the carboxylic acids, the stronger the acid is

- Trichloroethanoic acid is the **strongest acid** as:
 - The O-H bond in CCl3COOH is the **weakest** since there are **three** very strong electronegative Cl atoms withdrawing electron density from the -COOH group
 - When the O-H is broken to form the carboxylate (-COO^-) ion, the charge density is further spread out by the three electron-withdrawing Cl atoms
 - The carboxylate ion is so **stabilised** that it is less attracted to H^+ ions

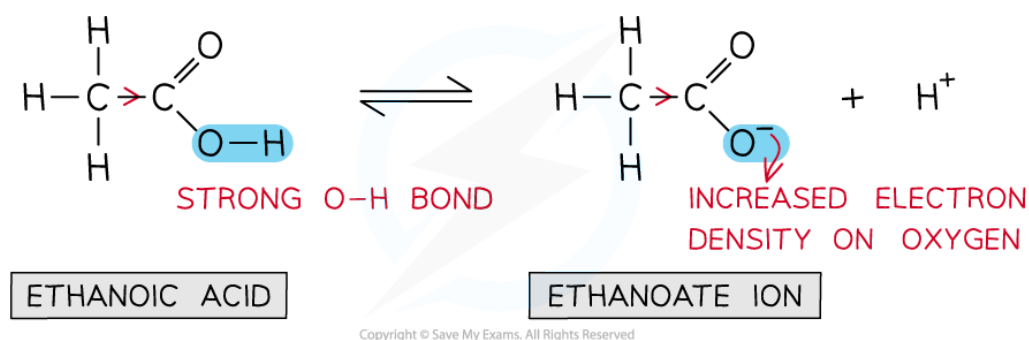
The equilibrium of trichloroethanoic acid and the trichloroethanoate ion



Relative acidity of trichloroethanoic acid

- Ethanoic acid is the **weakest acid** as:
 - It contains an **electron-donating** methyl group which **strengthens** the O-H bond
 - The methyl group **donates** negative charge towards the $-COO^-$ group which becomes more likely to accept an H^+ ion

The equilibrium of ethanoic acid and the ethanoate ion



Relative acidity of ethanoic acid